# Editorial Summary - MS1 · P14 AuroraCast

14.10.2026 · Reviews received: A, B

## What the reviewers and we see

AuroraCast forecasts the planetary Kp index for the next 3-hour interval, plus the probability of a storm (Kp ≥ 5), from live L1 solar-wind data, scored against GFZ Kp with persistence as baseline. Both reviewers and we single out the leakage section (GFZ delay, nowcast vs definitive Kp, L1-to-Earth shift) and the quantified data plan as exemplary. The central issue is scope and timing: four self-hosted services on a VM that is not chosen yet, while the live feed only keeps about 24 hours of data.

## Points to act on

Answer every point below in `docs/ms2_summary.pdf`, one numbered entry per point with a status (Changed / Partly / Rejected / Not yet) and a commit or file. Use these labels.

**Must**

- **A4 · Self-hosted stack is heavy for a solo core** - Airflow, Postgres/TimescaleDB, MLflow and Streamlit in Docker Compose on a cloud VM are four services to keep alive, and the provider is still open. Confirmed; this is our main concern too. Decide the cloud provider now and keep the core on the simplest host that works (your GitHub Actions fallback is a legitimate core, not a downgrade); move the rest to stretch until the pipelines run.
- **E1 · Start the poller now** - The live RTSW endpoint keeps only ~24 h, so every day without the poller is live data you cannot recover; OMNI fills history later, but processed and time-shifted, which is exactly the skew in A3. Run the hourly poller this week (on GitHub Actions if the VM is not ready) and write to persistent storage from the first run.

**Should**

- **A1 · Persistence baseline unmeasured and possibly weak** (also B3) - Persistence is defined as the last Kp *published* before t; with the GFZ delay it can be more than one interval old, which makes "10 % below persistence" cheap. Your targets are also "estimates to be re-justified", which invites moving goalposts. In the first EDA, compute persistence both ways (last published vs last completed interval) and the storm base rate, pin the targets to the stronger baseline, and freeze targets and decision thresholds in the README before touching the test set.
- **A2 · Give NOAA's forecast a target** (also B4) - NOAA SWPC's own Kp forecast is logged only as a reference. Make it part of the success criterion for the live period, e.g. storm recall at equal precision vs NOAA, shown on the dashboard. We do not see an originality problem in NOAA publishing a forecast; it simply makes it the right bar.
- **A3 · OMNI vs real-time skew is not quantified** (also B1) - Test years use quality-controlled OMNI data, the live system uses raw RTSW data and nowcast Kp; the proposal calls this a "small, known difference" without a number, so the offline MAE may overstate live performance. Once the poller has a few weeks of data, compare RTSW with OMNI for the same period (and nowcast vs definitive Kp) and report the gap.
- **B2 · Few independent storms in the test years** - Storms are a few percent of intervals and cluster, so recall at precision ≥ 0.40 on 2024-2025 will be noisy. State the number of independent storm events in the test set and report per-event hit rate with that count.

**Could**

- **E2 · Test the publication-delay rule** - Implement "which Kp is published before t" as one tested function in the shared feature module.
- **E3 · Proposal file** - The PDF has no text layer ("Print to PDF") and sits at `docs/Proposal_auroracast_final.pdf`; export it with text and commit it as `docs/proposal.pdf`.

## Looking ahead to MS2

MS2 is the feature pipeline: we expect the poller running on schedule since early October, the OMNI backfill in the store, the EDA with both persistence variants and the frozen targets, all on a host you have actually decided on.

## Feedback on the reviews you wrote

Both reviews are excellent and ground every critique in specific metrics, features or rules from the proposals. On P08 you identified the seasonal gap between the late-summer baseline measurement and the autumn evaluation, with solid fixes (walk-forward evaluation, trend adjustment, bootstrap CIs). On P11 you flagged an in-sample champion/challenger setup and a seasonally blind 8 % retrain trigger, with clear MS2 solutions. The only weakness is an overstated claim about test-set tuning on P11.